JianghuSect 2 (`JS2-v2.6.0`) — Deep-Dive Feature & Logic Audit

System #02 (Part 2 of 2): NPC Brain & Decision Architecture — Orphaned Game Systems, Hardcoded Exploration Ceilings, Utility/Cache Bugs & Goal/Petition State-Machine Defects

- **Target Codebase**: `Js2 new` (`SIM.PATCH_VERSION = '2.6.0'`, `SIM.SAVE_SCHEMA_VERSION = '2.6.0-living-chronicle-and-reporting'`)
- **Target Readers**: Autonomous Coding Models (**GPT** & **Gemini** Systems Architects / AI Engineers)
- **Prerequisite Reading**: `02A_JS2_Deep_Audit_NPC_Brain_Part1_Epistemic_Knowledge_and_Information_Asymmetry` (4-Tier Epistemic Matrix, God-View Leaks, Subjective Breakthrough/Power Estimation, and O(1) Social Gossip Sync).
- **Core Modules Audited**: `js/ai.js`, `js/cognition.js`, `js/cpeco.js`, `js/explore.js`, `js/content.js`, `js/cultivation.js`, `js/economy.js`, `js/sect.js`
- **Audit Scope (Part 2 of 2 — Action Coverage, Utility Math & Goal Loops)**:
  - **Section 5 (`[BRAIN-ORPHAN-05]`)**: Complete Catalog of **Orphaned & Hard-Capped Systems** That Autonomous NPCs Never Use or Cannot Reach (Pagoda Floors 10–43, Apex Trials, 4th Sanctum, All 4 Micro-Locations, Voluntary Vows, Elemental Rune Socketing, and Disembodied `content.js` Cron Loops).
  - **Section 6 (`[BRAIN-UTIL-06]`)**: Utility Scoring, Cache Invalidation & RNG-Purity Bugs (`A.study` Empty-Desk Loop, Raid-Blind 60-Tick `_cogModCtx` Cache, `W.rng` Mutation Inside `pickBasic`, and Top-6 Softmax Starvation Across 38 Actions).
  - **Section 7 (`[BRAIN-GOAL-07]`)**: Aspiration, Goal, Petition & Loyalty-Paradox Defects (100% Unread `W.sect.petitions` Queue, Overweight `none` Goal Suppressing Purposeful Agency, Missing Aspirations/Modifiers, and Editor-Cheat Instant Manual Mastery Rewarding Disloyalty).

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5. [BRAIN-ORPHAN-05] Action-Space Orphans — Systems Built in Code That Autonomous NPCs Never Reach or Weight

In a game where **the player cannot command NPCs**, any feature without a live, weighted path through `AI.ACTS` / `COG` / `EXPLORE.tryVolunteer` effectively **does not exist in real gameplay**, even if 100% of direct-call unit tests pass.

5.1 Complete Matrix of Orphaned or Artificial-Ceiling Systems

Sub-ID	Feature / Subsystem	Exact Code Location	Why Autonomous NPCs Never Reach It	Chain-Reaction Impact
5A	**Nine-Dao Pagoda Floors 10–43**	`js/explore.js:742` (`EX.tryVolunteer`)	Hardcodes `EX.departNineDaoPagoda(W, party, { maxFloor: 9 })` despite `EX.NINE_DAO_PAGODA.floors` having **43 floors** (`lines 553–597`)!	**34 out of 43 Grand Dao Paths** (Floors 10–43: `CRAFT_CREATION` .. `ARENA_GLORY`) **never** unlock their `RES_NINE_DAO_<PATH>` 3-piece resonance sets, `daoInsights`, or Pagoda `MANUAL_PATH_*` drops in natural play!
5B	**43 Apex Dao Trials (`attemptApexTrial`)**	`js/explore.js:542`, `js/cultivation.js`	Zero callers in `ai.js`, `cognition.js`, or `explore.js` (`tryVolunteer`). Only called in `v210` test suite and UI!	Stage 7–9 Grandmasters **never** challenge their path's Apex Trial, never earn `apexDaoTitle` (`+8% ATK/MATK,`

Sub-ID	Feature / Subsystem	Exact Code Location	Why Autonomous NPCs Never Reach It	Chain-Reaction Impact
				`+6% DEF`, `+12% cultRate`), and never mint first-clear `CRYSTAL_PATH_*`.
5C	**4th Ancient Sanctum (`SANCTUM_MYRIAD_BEAST_TOTEM`) & 31-Path Bonus Pool**	`js/explore.js:737–741` & `422, 440`	`tryVolunteer` maps `BODY -> BLOOD_VAJRA`, `GOD/DEMON -> ASTRAL_DIVINE`, and **all 40 other paths -> `SWORD_ABYSS`**, completely omitting `SANCTUM_MYRIAD_BEAST_TOTEM`! Line 440 also excludes `t.grandDaoLegendary`.	`SANCTUM_MYRIAD_BEAST_TOTEM` has **0 autonomous runs**; Beast/Sonic/Zither/Whip disciples never farm their themed sanctum; 31 newer paths get off-theme bonus manuals.
5D	**All 4 Living Micro-Locations (`EXPLORE.MICRO_LOCATIONS`)**	`js/explore.js:807–980` (`dispatchExploration`)	`dispatchExploration` has **zero callers** outside unit tests and UI buttons!	Sword Tomb (`+SWORD`), Ancient Mech Ruins (`+FORGING`), Myriad Venom Forest (`clearToxin`), Moon Mirror Valley (`clearHeartDemon`), and regional Outpost

Sub-ID	Feature / Subsystem	Exact Code Location	Why Autonomous NPCs Never Reach It	Chain-Reaction Impact
				camp perks (`applyCampPerksToParty`) are **100% dead** in autonomous runs!
5E	**Voluntary Dao Vows (`swearVow` & `redeemVow`)**	`js/cultivation.js` (`VOLUNTARY_VOWS`)	Zero utility/cognition logic ever calls `swearVow` or `redeemVow`.	Ascetic monks, proud duelists, and remorseful vow-breakers never organically swear `VOW_NO_MEDICINE` / `VOW_NO_RETREAT` / `VOW_ASCETIC_PRECEPT` or redeem broken vows (`EXP_TRAIT_OATH_KEEPER` starved).
5F	**Elemental Rune Socketing (`ECO.socketElementalRunes`)**	`js/economy.js`	Never called by `A.forge`, `COG`, or `SECT.runOfficerJurisdictionSingle('forge')`.	Weapons/armor never gain socketed elemental runes autonomously, starving `EX.solveElementalPuzzleGate` (`explore.js:769`) of

Sub-ID	Feature / Subsystem	Exact Code Location	Why Autonomous NPCs Never Reach It	Chain-Reaction Impact
				dual/triple elemental keys!
5G	**Research Experiments & Organ Refinement as Real Activities**	`js/content.js:510-530` (`C.daily`)	Executed via disembodied midnight cron `W.npcs.find(...)` every 5/7 days instead of `AI.ACTS` + `COG` goals!	Ignores physical stations, duration, sleep/seclusion, and allocates precious Rare Beast Organs in `W.npcs[0]` array-index order instead of merit, officer triage, or personal craft ambition!

5.2 Architectural Remediation Blueprint for AI Developers

1. **Unblock All 43 Pagoda Floors & Autonomous Apex Trials (`js/explore.js`)**:
   - In `EX.tryVolunteer`, replace `{ maxFloor: 9 }` with dynamic progressive targeting: e.g., target the party's own `cult.path` floor (`EX.NINE_DAO_PAGODA.floors.find(f => f.path === lead.cult.path)?.floor`) or climb sequentially from `(W.world.nineDaoPagoda.highestCleared || 0) + 6` up to `43` scaled by party `CULT.power`!
   - When an expedition clears a Pagoda floor matching an `equivalentStage >= 7` party member's `cult.path` (and `!member.apexDaoTitle`), automatically trigger `EX.NINE_DAO_PAGODA.attemptApexTrial(W, member, floor.floor)`!
2. **Route All 4 Sanctums & All 4 Micro-Locations Through Autonomous Exploration (`js/explore.js`)**:
   - Score all 4 `EX.SANCTUMS` by party subpath/weapon/element fit (so `BEAST_SOVEREIGN`, `SONIC_HARMONY`, `VENOM_GU`, `PRIMORDIAL_SHIFTER`, `SPIRIT_HUSBANDRY` naturally pick `SANCTUM_MYRIAD_BEAST_TOTEM`!) and include all 43 paths in `rollSanctumBonusReward`.

    - Wire `MICRO_LOCATIONS` into `EX.tryVolunteer` (choosing the micro-location & `choiceId` that matches the volunteer's needs—e.g., `MEDITATE_MOON_REFLECTION` when `heartDemon >= 35`, `SUBDUE_GU_INSECT` when `toxic >= 30`, `COMPREHEND_SWORD_INTENT` for Sword practitioners, `DECIPHER_PUPPET_BLUEPRINT` for `MECH_PUPPET` / `FORGING`!).
3. **Add Genuine Cognitive Hooks for Voluntary Vows, Rune Socketing & Physical Research (`AI.ACTS.experiment` / `refineOrgan`)**:
    - During `COG.prepare` (monthly review or major life milestone): high-`discipline` + high-`integrity` monks/ascetics autonomously swear `VOW_ASCETIC_PRECEPT` or `VOW_NO_MEDICINE`; high-`courage` + high-`pride` vanguards swear `VOW_NO_RETREAT`; remorseful disciples with `violated === true` and `merit >= 60` pursue `redeemVow`.
    - Extend `A.forge` (when `FORGING >= 35` or `ARRAY >= 35`) to socket surplus elemental materials via `ECO.socketElementalRunes` on un-socketed T3+ equipment, and convert `C.runExperiment` / `C.refineOrgan` from `C.daily` instant telepathy into physical `alchemy` / `forge` / `infirmary` station actions!

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6. `[BRAIN-UTIL-06]` Utility Scoring, Cache Staleness & RNG-Purity Bugs

6.1 Exact Code Anchors

- `js/ai.js`: `A.study` (lines 495–520), `A.guard` (lines 599–611), `A.heal` (lines 612–620), `AI.decide` (lines 830–875)
- `js/cognition.js`: `COG.modifiers` `_cogModCtx` cache (lines 659–700)
- `js/cpeco.js`: `pickBasic` (lines 110–126)

6.2 Root-Cause Code & Mathematical Defects

Defect 6A — `A.study.can = () => true` Causes Aimless "Blank-Desk Reading" & Scripture Station Starvation (`ai.js:496`)

`A.study` sets `can: (n, W) => true`.

- When `!n.cult` (a mortal who should be running `study_dao`, not `study`), OR when `!!n.cult && !n.cult.learning && !AI.nextTechToStudy(W, n)` (zero affordable/eligible unlearned techniques exist in `W.sect.library`), high-`curiosity` / high-`scholarship` NPCs still score `+0.70..+1.15` on `A.study`!

- `SA.start` (`cpeco.js:360`) falls back to `oStart` (`ai.js:507`), which returns `true` even with `n.cult.learning === null`.
- **Ripple Effect**: Scholars sit inside `scripture` for 100–150 minutes reading nothing, occupying scarce `'read'` stations and blocking mortals (`study_dao` -> `blocked`) and disciples who actually saved CP to learn a new manual (`COG.blocked` -> `scripture` goal expires!).
- **Fix**: Gate `A.study.can` on `!!n.cult && (!!n.cult.learning || !!AI.nextTechToStudy(W, n) || ((n.h.skills.SCRIPTURE || 0) < 60 && (n.post === 'scripture' || n.assist === 'scripture' || n.cult.path === 'LORE_ARCHIVIST' || n.cult.path === 'INK_LITERATI')))`. Similarly gate `A.heal.can` on `n.h.state.hp < CULT.hpMax(n) * 0.88 || (n.inj && n.inj.length > 0)` and `A.guard.can` on actual guard eligibility instead of returning `true` and relying on `-3` scores!

Defect 6B — 60-Tick Cache Blindness in `COG.modifiers` Ignores Sect Raids, Alert Shifts & Weather Changes (`cognition.js:666`)

Inspect `n._cogModCtx` invalidation in `cognition.js:666`:

- `mc` is cached for up to **60 ticks (1 in-game hour)**.
- **Missing Cache Keys**:
  1. `W.threat.raid` and `W.threat.alert`: `roleNeed(W, n)` returns `1.0` during a raid vs `0.15` in peacetime (`dutyActVal` jumps by `+0.50`+!). Because `W.threat.raid` and `W.threat.alert` are **omitted** from the invalidation check on line 666, when a raid or alert triggers mid-hour, Commanders, Vanguards, and Guards keep their peacetime `mc` for up to 59 minutes!
  2. `W.weather.kind`: `mc.weatherPen` is cached without checking `mc.weather !== (W.weather && W.weather.kind)`.
  3. `s.heartDemon` and `n.cult.toxic` thresholds (`45`): checked live at lines 702–704, **except** `mc.hp` is stored at line 691 and never checked on line 666!
- **Fix**: Include `mc.raid !== !!(W.threat && W.threat.raid) || mc.alert !== ((W.threat && W.threat.alert) || 0) || mc.weather !== (W.weather && W.weather.kind)` in the `_cogModCtx` invalidation check (all O(1) primitives!).

Defect 6C — `W.rng.next()` Consumed Inside `pickBasic(W, n)` Breaks Read/Score Purity (`cpeco.js:110–126`)

```javascript
function pickBasic(W, n) {
  if (W && W.t > 0 && n._pbTick === W.t) return n._pbVal;
  ...
```

```
const s = (tgt - have) / tgt + W.rng.next() * 0.1 - r.tier * 0.05;
```

- Calling `A.brewBasic.can(n, W)` or `A.brewBasic.score(n, W, S)` consumes `W.rng.next()` AND mutates `n._pbTick` / `n._pbVal`!
- At `W.t === 0` (`W.t > 0` is `false`!), caching is bypassed completely, so `can()`, `score()`, `_buildWhy()`, and `where()` **each roll `W.rng.next()` separately** and can pick different recipes! Even at `W.t > 0`, if any UI/diagnostic inspector evaluates action scores on a tick where `n` was mid-action, it mutates `W.rng` and `n._pbVal`, violating **Hard Invariant #3 (Zero RNG / Zero State Mutation on Read/Score)**!
- **Fix**: Replace `W.rng.next() * 0.1` in `pickBasic` with a pure deterministic tie-breaker `(((U.hash32 ? U.hash32(n.id + ':' + r.id + ':' + W.day) : (n.n + i * 7 + W.day)) % 100) / 1000)`—zero `W.rng` calls, 100% pure!

Defect 6D — Hard `top = cands.slice(0, 6)` Softmax Cutoff Across ~38 Actions (`ai.js:861`)

With ~38 registered actions in `AI.ACTS`, slicing to `6` before softmax permanently zeroes out #7+ even when scores are tied within `0.02`. Widen the softmax candidate window to `top = cands.filter(c => c.u >= mx - Math.max(0.65, temp * 3.5)).slice(0, 12)` so contextually competitive niche actions (fishing, clue investigation, flatter, shareMedicine, experiment, path rituals) remain reachable according to `choiceTemperature(n)`.

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7. [BRAIN-GOAL-07] Aspiration, Multi-Step Goal, Dead-Petition & Loyalty-Paradox Defects

7.1 Exact Code Anchors

- `js/cognition.js`:
  - `ASPIRATIONS` & `COG.evaluateAspirations` (lines 35–148)
  - `COG.prepare` personal manual cheat & goal weights (lines 515–525, 580–652)
  - `COG.modifiers` aspiration bonus (lines 695–699)
  - `COG.submitPetition` (lines 1005–1022)

7.2 Root-Cause Code & Logic Defects

Defect 7A — `W.sect.petitions` Is a 100% Unread Dead Queue (`cognition.js:1005–1022`)

When corrupt welfare allocation passes over an injured high-merit disciple (`cpeco.js:351`), the disciple files an official complaint via `COG.submitPetition(W, p.n, 'complaint', { accusedId: tr.id, ... })` into `W.sect.petitions`.

- **Broken Loop**: **Zero functions in `cognition.js` or `sect.js` ever read `W.sect.petitions`!** Complaints against corrupt officers, medicine petitions, and equipment petitions sit unread forever until evicted at length 16.
- **Fix**: Wire `W.sect.petitions` resolution into `COG.performDuty` for:
  - `'discipline'` & `'censor'`: review open `'complaint'` petitions -> increment `W.sect.suspicion`, audit `accusedId` (`exposeEmbezzle` or reprimand biased allocation), mark petition `'resolved'` (restoring complainant `sat` / `trust`) or `'dismissed'` (if corrupt officer covers up for a friend, increasing `undercurrent`!).
  - `'healer'` / `'treasury'`: fulfill open `'medicine'` petitions from emergency reserves.
  - `'forge'`: queue craft/catalyst orders for open `'equipment'` petitions.

Defect 7B — Overweight `'none'` Option (`1.2 + freedom/100 = 1.70`) Suppresses Goal Formation (`cognition.js:642`)

In `COG.prepare`, real goals (`scripture`, `gear_upgrade`, `help_friend`, `mentor_disciple`, `investigate_clue`) typically score **`0.65`** to **`1.15`**, whereas line 642 pushes `{ id: 'none', weight: 1.2 + E(n, 'values.freedom') / 100 }` (**`1.40`** to **`2.20`**)!

- When `'none'` wins, `c.goal` stays `null` for 180 ticks (`3` hours). Even driven disciples (`ambition = 80, discipline = 80`) frequently roll `'none'` and spend half their lives without an active multi-step goal!
- **Fix**: Scale `'none'` weight down to `clamp(0.35 + (E(n, 'values.freedom') - E(n, 'personality.ambition') * 0.6 - E(n, 'personality.discipline') * 0.5) / 140, 0.10, 0.95)` so carefree (`freedom = 90, ambition = 20`) spirits often drift freely while ambitious, disciplined cultivators almost always maintain an active goal!

Defect 7C — Incomplete Aspiration Coverage & Missing `COG.modifiers` Hooks (`cognition.js:35–148, 695–699`)

1. `wpnSk` (`line 100`) omits `ROSARY`, `CLAW`, `NEEDLE`, `FAN`, `ASTROLABE`, `ABACUS`.

2. Missing aspirations for **Spirit Beast Master** (`beast_sovereign`), **World Explorer / Seeker** (`horizon_seeker`), and **Shadow / Unorthodox Powerbroker** (`shadow_schemer`).
3. `COG.modifiers` (`lines 696–698`) grants action utility bonuses for **only 3 of the 9 aspirations** (`peaceful_companion`, `martial_master`, `benevolent_healer`), completely forgetting `realm_ascendant` (`cultivate`/`consolidate`/`paidTraining`), `master_alchemist` (`brew`/`personalBrew`/`experiment`), `master_artificer` (`forge`/`experiment`), `respected_mentor` (`pair`/`guidedStudy`), `wealthy_trader` (`outsideTrade`/`forage`/`mine`), and `sect_pillar` (`duty`/`guard`)!

Defect 7D — Personal Manual Instant-Mastery Cheat & The Loyalty Punishment Paradox (`cognition.js:516–525`)

```javascript
if (tObj && (tObj.reqTier || tObj.tier || 1) <= effSt && CULT.learnable(n,
[tid]).length > 0 &&
    (W.sect.library.includes(tid) || E(n, 'personality.loyalty') < 52)) {
  SECT.EDITOR.usePersonalItem(W, n.id, 'TECH_MANUAL', tid);
```

1. **Sandbox Cheat Leak**: `COG.prepare` calls `SECT.EDITOR.usePersonalItem`, which **instantaneously adds `tid` to `n.cult.techs` in 0 ticks**, skipping `n.cult.learning` (`CULT.studyCost`) completely!
2. **Inverted Loyalty Incentive**:
    - A **low-loyalty (`< 52`)** disciple who finds or buys a manual masters a Tier-7 art **instantaneously in 0 ticks for 0 CP**!
    - A **high-loyalty (`>= 52`)** disciple who bought that manual with their own Spirit Stones is forbidden from reading their own scroll, forced to donate it for `0.55x` CP, and then charged `1.00x` CP +数百 ticks of `study`!

- **Fix**: Never call `SECT.EDITOR.usePersonalItem` from `COG.prepare`! Instead, allow **any** disciple holding a compatible personal `tid` in `n.inv.techs` (who wants to learn it—whether loyal or not!) to start `n.cult.learning = { id: tid, need: CULT.studyCost(tObj, n), got: 0, fromPersonalScroll: true }` for **0 CP fee** when entering `A.study`; once finished learning, a **loyal (`>= 55`)** disciple donates the physical scroll to `W.sect.library` for full `CPE.techReward` CP, whereas a selfish (`< 40`) disciple keeps it private or sells it! That makes loyalty **win-win** (you read the scroll you found/bought *and* contribute it to the sect afterward!) while requiring real `study` time!

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8. Mandatory End-to-End Verification Suite Spec (For System #02 Part 2)

Add `test/v270_npc_brain_part2_coverage_and_goals_suite.js` asserting:

1. **`test_autonomous_pagoda_climbs_past_floor9_and_triggers_apex_trial`**: Run autonomous `tryVolunteer` with a Stage-8 `CHRONO_EPOCH` party; verify `departNineDaoPagoda` reaches Floor 24+ and triggers `attemptApexTrial`, awarding `apexDaoTitle`.
2. **`test_all_4_sanctums_and_micro_locations_visited_autonomously`**: Verify autonomous exploration dispatches parties to `SANCTUM_MYRIAD_BEAST_TOTEM` and `EXPLORE.MICRO_LOCATIONS` based on path/state needs (e.g., high `heartDemon` -> `LOC_MOON_MIRROR_VALLEY`).
3. **`test_pickBasic_and_scoring_zero_rng_purity_at_tick0_and_tickN`**: Assert `A.brewBasic.can` and `A.brewBasic.score` consume `0` `W.rng` calls at both `W.t === 0` and `W.t = 500`.
4. **`test_cog_modifiers_cache_invalidates_immediately_on_raid`**: Populate `_cogModCtx` in peacetime, set `W.threat.raid = true` on the next tick (`W.t + 1`), and verify `commander` / `guard` `COG.modifiers(W, n, 'duty')` immediately reflects raid urgency without waiting 60 ticks.
5. **`test_petitions_adjudicated_by_discipline_and_censor_officers`**: Submit an open `'complaint'` petition against a skimming treasurer; advance `SIM.tick` until the `discipline` elder completes `A.duty`, and verify the petition transitions out of `'open'` and triggers investigation/suspicion handling.
6. **`test_personal_manual_requires_real_study_and_rewards_loyal_sharing`**: Give a loyal (`loyalty = 80`) disciple a compatible unowned `tid` in `n.inv.techs`; verify they start `c.learning` for `0` CP (not instant 0-tick `EDITOR` mastery) and donate the scroll to `W.sect.library` after learning.